

10.	B	COE: gain in GPE by P + gain in KE by P and Q = Loss of GPE by Q $4 \times 9.81 \times 3\sin 30^{\circ} + \frac{1}{2} (6+4) v^2 = 6 \times 9.81 \times 3\sin 40^{\circ}$ $v^2 = 10.9 \quad v = 3.31 \text{ m s}^{-1}$
11	C	At constant speed